COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Mansgement	**SECTION:**	CPE31S2
DATE PERFORMED:	13/08/2026	**DATE SUBMITTED:**	13/08/2026
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PRELIM EXAM SKILLS

1. PROCEDURE OUTPUT

Set up:

Hostname	Operating System	IP Address
queleroyubuntu3@server1	Ubuntu	192.168.56.9
queleroyubuntu3@server2	Ubuntu	192.168.56.10
leroyq@localhost	CentOS	192.168.56.12

Part A: Installing and Verifying Ansible

1. Update the package repository.
sudo apt update

2. Install pip if it is not yet installed.
sudo apt install python3-pip -y

3. Install Ansible.
python3 -m pip install --user ansible
or
sudo apt install ansible

4. Verify that Ansible is installed correctly.
ansible --version

5. Verify SSH connectivity to each managed node.
ssh student@<ubuntu1-ip>
ssh student@<ubuntu2-ip>
ssh student@<centos1-ip>

Submit a screenshot of the following:
- Ansible version
- SSH connectivity

Ansible version (OUTPUT):
Ubuntu (Control Node):

```
queleroyubuntu3@workstation:~$ ansible --version
ansible 2.10.8
  config file = None
  configured module search path = ['/home/queleroyubuntu3/.ansible/plugins/modul
es', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0]
```

SSH connectivity (OUTPUT):
Control node to ubuntu1:

```
queleroyubuntu3@workstation:~$ ssh queleroyubuntu3@192.168.56.9
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jul 30 09:20:53 2026 from 192.168.56.11
queleroyubuntu3@server1:~$
```

Control node to ubuntu2:

```
queleroyubuntu3@workstation:~$ ssh queleroyubuntu3@192.168.56.10
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jul 30 09:26:51 2026 from 192.168.56.11
queleroyubuntu3@server2:~$
```

Part B.

```
queleroyubuntu3@workstation:~/ansible-lab$ ansible all -m ping -i inventory.yaml
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

Part C.

```
queleroyubuntu3@workstation:~/ansible-lab$ ansible linuxservers -m ping -i inventory.yaml
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

Part D.

```
queleroyubuntu3@workstation:~/ansible-lab$ cat inventory.yaml
all:
  children:
    ubuntu:
      vars:
        ansible_user: queleroyubuntu3
      hosts:
        ubuntu1:
          ansible_host: 192.168.56.9
          os_type: Ubuntu
        ubuntu2:
          ansible_host: 192.168.56.10
          os_type: Ubuntu

    linuxservers:
      children:
        ubuntu:
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance

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